CHE-494: Biopharmaceutical Process Development and Manufacturing

Project Assignment

Guidelines:

1. **Please do not share and discuss your assignment with others.**
2. **Two case studies italicized to tailor your answer accordingly.**
3. **Prepare a slide presentation to answer and communicate with the class.**
4. **You may please work on the questions during the course of instructions as and when pertinent classes are delivered. Hence there is no separate homework or written quizzes.**
5. **Questions cover all the course contents.**

<u>Questions:</u>

1. What makes you excited (or not) to work for biopharmaceutical company?
2. *What is the most common purification technique for the large molecule drug substance? Provide key considerations for the process development and reliable scale-up of the same.*
3. What are the key considerations while selecting the crystal form of the small molecule drug substance? Why?
4. What is the most challenging small molecule drug product process? What are the challenges and how to make this drug product process work?
5. What is the simplest small molecule drug product process? What are the challenges to make this drug product process work?
6. What would be your considerations while designing and scaling-up of mixing processes for biologics molecules? Why?
7. *<u>A poorly soluble small molecule drug substance</u> (active pharmaceutical ingredient, API) has <u>very bad physical properties</u>, namely the broader particle size distribution, poor flowability and low bulk density. What drug product process would you recommend for solid dosage form? Provide a process flow chart and outline process design considerations for the unit operations involved.*
8. What is a common drying process used for the large molecule drug product? Provide key considerations for the process development and reliable scale-up of the same.